VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

VISHWAKARMA AWARDS 2026-27

Vishwakarma Awards 2026–27

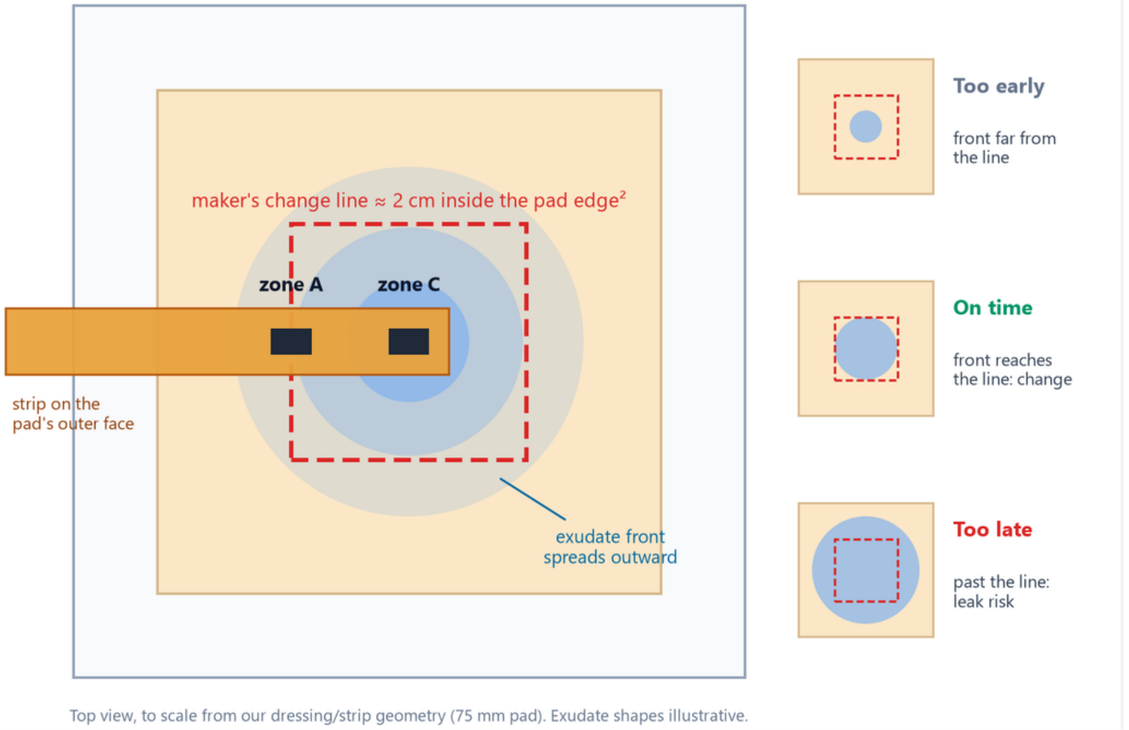
VK26-IITI-HT-0050

Team: Soham Prabhakar Kadu · Bhakti Ajay Kadam

Problem Context

EVIDENCE FROM PUBLISHED SOURCES

44.9 %	≈ 2 cm · 75 %	1 point	> 10×
DRESSING CHANGES MADE WHILE MOISTURE WAS OPTIMAL[1]	SOME DRESSINGS SHOW WHEN IT IS TIME TO CHANGE[2][14]	ONE EXISTING SENSOR TAKES A SINGLE-POINT READING[1]	REPORTED EXUDATE AMOUNTS VARY BY >10×[6]



Top view, to scale from our dressing/strip geometry (75 mm pad). Exudate shapes illustrative.

Our illustration shows why the change line can be hidden under a cover (drawn to scale from our strip design).

The problem and who we designed for

- **Problem:** a carer cannot see whether a covered foam dressing has reached its maker's change indicator.
- **Result:** changing too early can cause discomfort and waste; changing too late can lead to leakage and wet skin.
- **Who:** nurses and family carers supporting people with draining chronic wounds, including diabetic foot, leg and pressure ulcers.
- **We will:** show that change indicator from outside, without lifting the dressing.

Project at a glance: Dermasense · Team VK26-IITI-HT-0050 · Health Tech: Emergency Response & Remote Monitoring · Our idea is to read a dressing's change indicator from outside, without lifting it.

What carers can use today — and what is still missing

EARLY SEARCH · NOT EXHAUSTIVE

Current options and their limits

Today's option	Without lifting?	Maker's rule?	What limits it
Fixed schedule (clock)	✗	✗	hard to predict because fluid amounts vary >10×[6]
Lift and look	✗	✓	requires removing the dressing; removal can hurt[20]
Visual change indicator[14]	✗	✓	only works when the indicator is visible
Single-point moisture sensor[1]	◐	✗	measures moisture at one point
Closed-loop smart bandage[5]	✓	✗	tested in mice; also delivers treatment[5]
Diaper wetness strip[4]	✓	✗	designed for diapers, not wound dressings
Dermasense (proposed)	✓	✓	**not yet tested; E2 and E4 will evaluate it**

✓ yes ✗ no ◐ partly. Our search of papers, patents and products was not exhaustive.

What we searched and what we are testing

- **Searched:** papers, patents and products[14514].
- **Found:** we found no low-cost sensor in our search that reads a dressing's own change indicator.
- **Our focus:** test whether the strip can show a carer when fluid reaches that indicator, without lifting the dressing.
- **Why now:** carbon can be laser-written on polyimide in one pass[7], and some dressings already use a visible change indicator[214].
- **Not our goal:** diagnosis or treatment; we are exploring a way to report how full the dressing is.

Question we are trying to answer: How can a carer tell when a covered dressing needs changing without lifting it?

User Journey: Before State

CURRENT CARE · PUBLISHED EVIDENCE

1.9–4.5 per 1,000	101 million	19–34 %	40.3 %
CHRONIC WOUND PREVALENCE IN TWO INDIAN SURVEYS[16][17]	PEOPLE IN INDIA LIVING WITH DIABETES[18]	ESTIMATED LIFETIME FOOT-ULCER RISK WITH DIABETES[19]	PATIENTS SAY DRESSING-CHANGE PAIN IS THE WORST PART[20]

Illustrative persona, not a real patient: a person with a diabetic foot ulcer living at home with family support and periodic nursing visits.

STAGE	1 Dressing is applied	2 Dressing fills over time	3 Carer decides	4 Change	5 Repeat
WHAT HAPPENS	A foam dressing sits under a bandage or sock	Fluid builds up beneath the cover	Follow a schedule, take a look, or wait for leakage	Too early or too late	About five visits a week[13]
PAIN POINT	The change indicator is covered	There is no clear signal	The decision is uncertain	Discomfort and waste, or leakage and wet skin	About 18 minutes per visit[13]

Illustrative care journey based on published data; the persona is not a real patient.

What published studies tell us

- **Time:** One UK patient survey reported about 18 minutes per dressing visit and five visits a week[13].
- **Waste:** In one study, 44.9% of recorded changes happened while moisture was still in the optimal range[1].
- **Cost:** Retail listings showed ₹240–510 for 10 × 10 cm foam dressings[15]; pack details are unclear. An early change may mean replacing a dressing sooner.
- **Safety:** if a dressing stays on too long it may leak and leave surrounding skin wet; removal can hurt[20].
- **Who carries it:** the patient, family carer and nurse all manage this routine.

Insight: when a dressing is covered, a carer may not be able to see whether it has reached its change indicator.

User Journey: After State (Intended)

DESIGN INTENT · NOT CLINICALLY TESTED

0 lifts	44.9 %	≤ ₹100	Not measured
DESIGN GOAL: CHECK WITHOUT LIFTING	OF RECORDED CHANGES WERE AT OPTIMAL MOISTURE[1]	TARGET STRIP COST (ESTIMATE)	COST SAVINGS HAVE NOT BEEN MEASURED

The same illustrative persona with the proposed workflow. This has not been clinically tested.

STAGE	1 Strip on	2 Reader on	3 Check status	4 Change indicator reached	5 Nearby check
WHAT HAPPENS	placed on the pad's outer face	clips on: 4 pins, 2 dowels	Green OK, without lifting the dressing	Amber CHANGE; red CHECK for a possible leak	A nearby phone may read the Bluetooth status; hardware test is pending
GAIN	nothing on the wound	same routine	A clearer decision, if testing supports it	A prompt to check the dressing	Local status only; no change-reason log

Intended workflow only. Bench testing comes first; the device has not been tested on people.

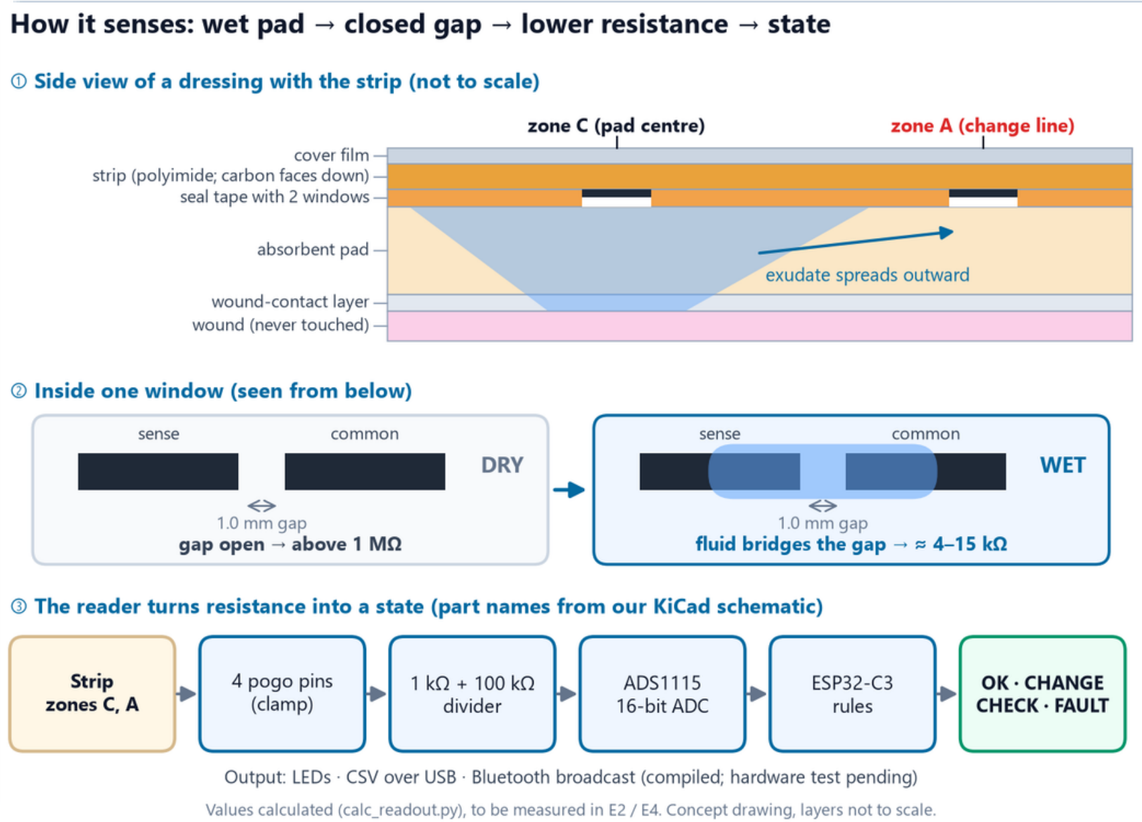
What each person could gain

- **Patient:** If validated, fewer unnecessary changes could mean less removal-related discomfort[20].
- **Family carer:** A simple status could help replace guesswork; the workflow still needs testing.
- **Nurse:** could check the status nearby without lifting the dressing. The current logger does not record why a dressing is changed.
- **Budget:** Target strip cost is ≤₹100 and the reader is reused. Any savings remain unproven.

Design Principle: The intended setup keeps the usual care routine. The strip sits on the dressing's outer face; the reader is reused. E4 will test the ±10% alert target.

System Architecture & Specs

DESIGN TARGETS · NOT YET BENCH-TESTED



Concept drawing: how a moisture reading could become a status. Values are design targets, not measured results.

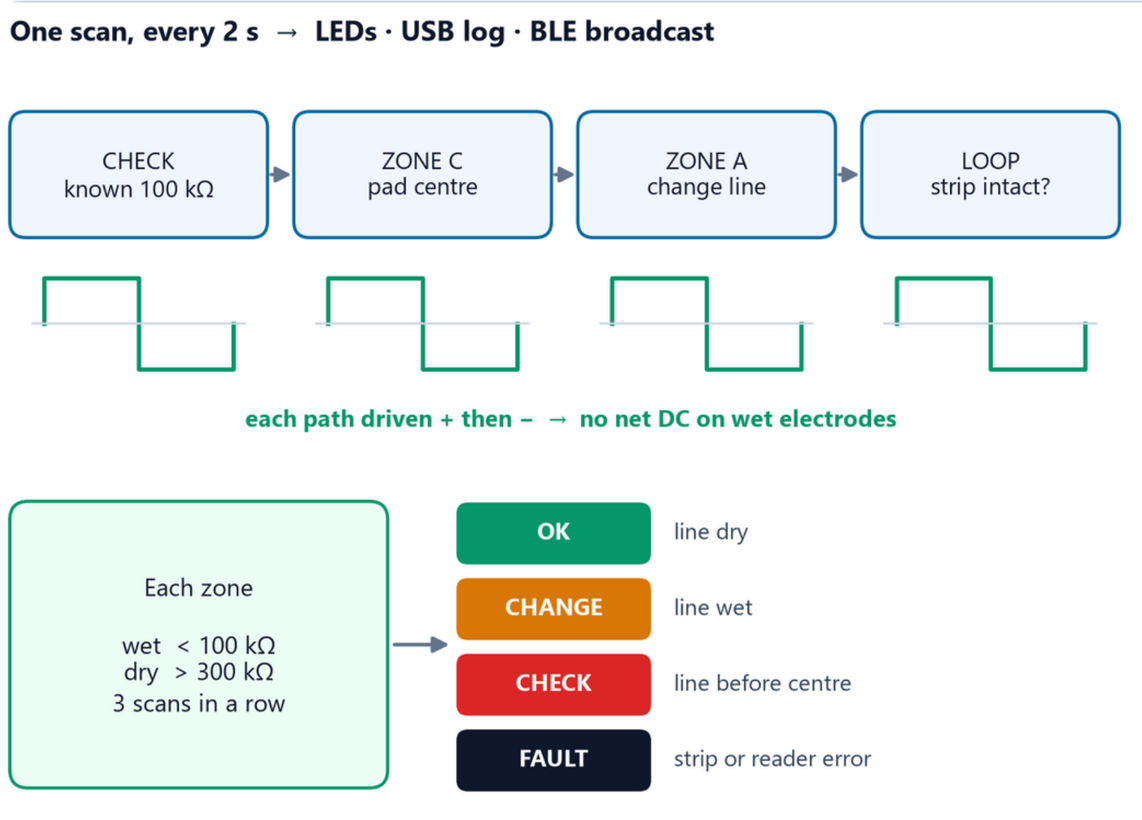
Design targets and hardware

Parameter	Design value	Basis
Sensing zones	2: centre C, change line A	decision D-012
Electrodes	1.0 mm lines and gaps	E2 will measure
Wet / dry	< 100 kΩ / > 300 kΩ, 3 scans	calc: wet 4–15 kΩ
Drive	bipolar, ≤ 33 µA, no net DC	calc · bench T8
ADC / MCU	ADS1115 16-bit (I²C) · ESP32-C3	datasheets
Self-test	100 kΩ check ±5 % + strip loop	show FAULT if the check fails
Contacts	4 pogo pins + 2 dowels	CAD checked
Output	LEDs · USB log · BLE broadcast	BLE compiled only

Architecture Rationale: We are designing the reader for reuse; the strip is the disposable part.

System Architecture & Specs: Firmware & Data Flow

SOFTWARE CHECKS PASSED · HARDWARE TESTING PENDING



This diagram follows one reader scan from measurement to status.

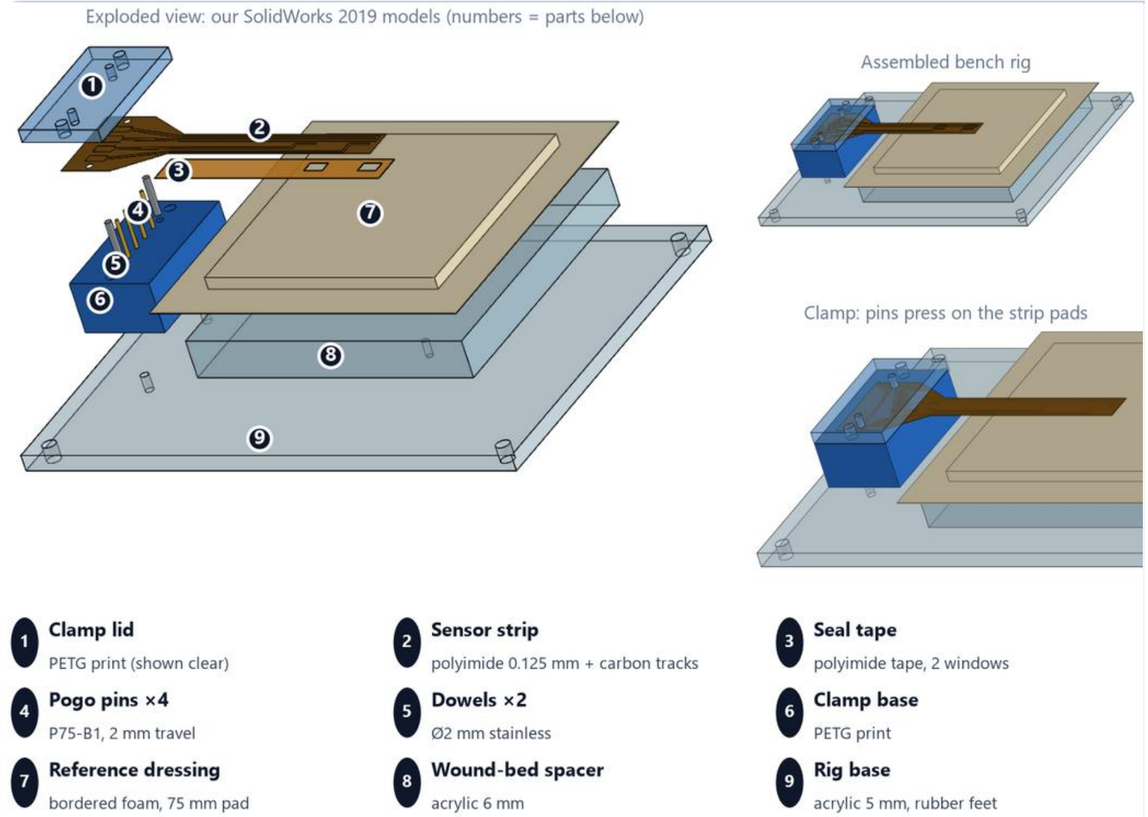
How the firmware turns readings into a status

- **Measure:** Each zone is driven in both directions to avoid net DC; hardware testing is pending.
- **Read:** The 16-bit ADC reading is scaled to the 3.3 V rail.
- **Decide:** The design thresholds are wet <100 kΩ and dry >300 kΩ for three consecutive scans.
- **Self-check:** A known 100 kΩ reference is checked each scan; a failed check shows FAULT.
- **Output:** Status appears on LEDs. Data can be logged over USB or read from a local Bluetooth broadcast; there is no custom app yet.
- **Why remote:** A nearby phone may read the status without lifting the dressing. This local Bluetooth workflow has not been tested on hardware; there is no remote monitoring.

Software checks: We passed 23 compile-time checks and 16 PC-side tests. The firmware has not yet run on a wired board.

CAD Design & Mechanical Specs

DESIGNED · CAD VERIFIED · NOT YET BUILT



Our SolidWorks models and drawings; the clamp has not been built yet.

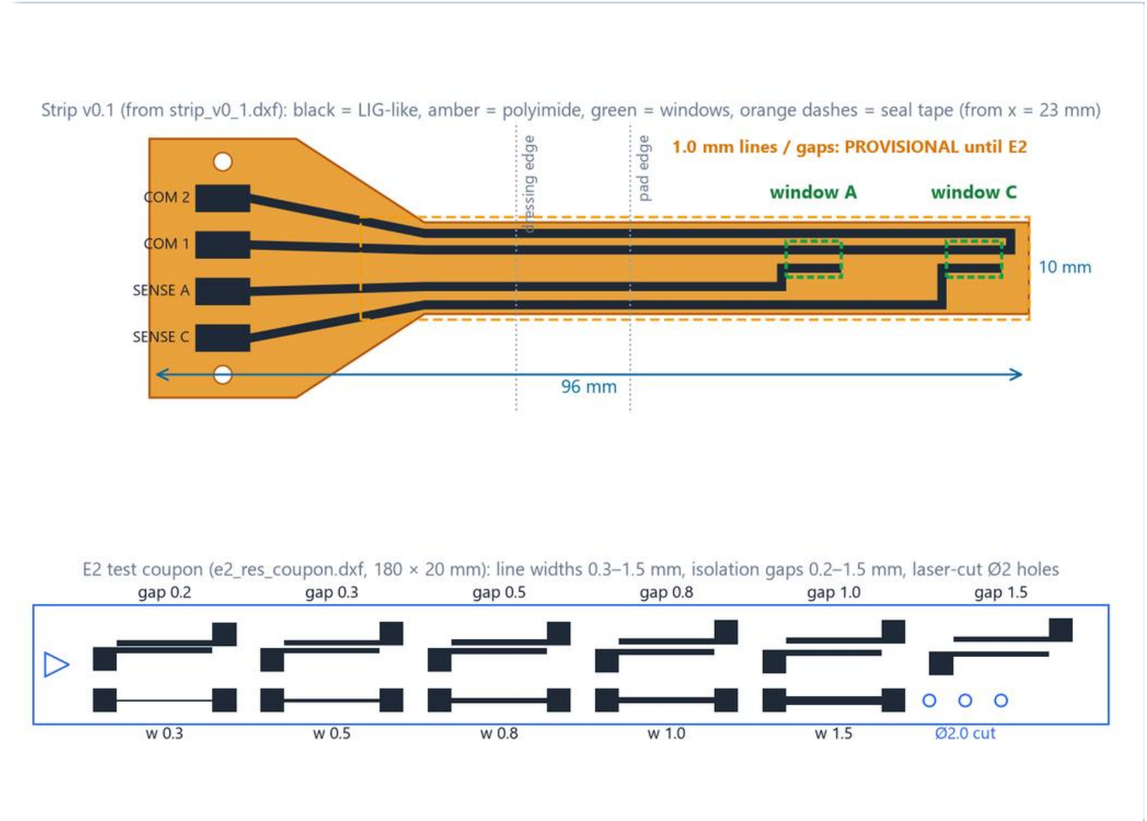
A clip-on clamp: 4 spring pins, 2 dowels

Check	Result	Basis
Materials	polyimide strip · PETG clamp · acrylic rig	printable, laser-cut
Pin force	0.72 N each · 2.9 N total	calc
Lid bending	11 µm vs 1.48 mm pin travel	calc
Alignment	0.49 mm used of ±1.35 mm	calc · 2.8× margin
Strip bending	0.62 % strain at 10 mm radius	calc · yield ≈ 3 %
CAD checks	99 passed · 0 failed · 5 blocked until E2 measurements	verify_cad.py
FEA	not run: loads are under 3 N	hand calcs above

Design Iteration: We found that the seal tape would sit under the clamp. We moved it to x = 23 mm; the CAD check now reports no clash (D-013).

CAD Design & Mechanical Specs: Sensing Strip & Laser Process

DESIGNED · LAB TEST PENDING



Laser pattern from our DXF files; it has not been fabricated yet.

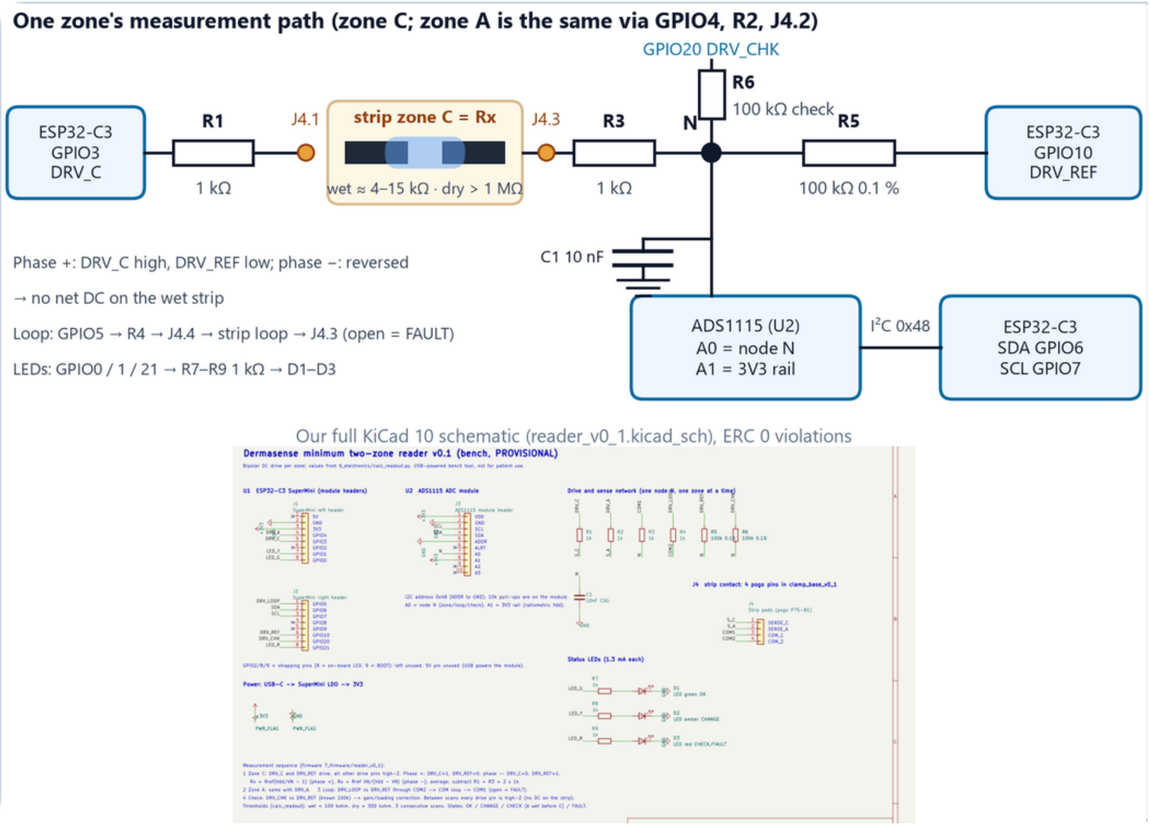
E2: laser and strip validation

- **Material:** We plan to create a laser-carbonised, LIG-like layer on polyimide[7].
- **Reference:** The machine's "12" setting is likely a PWM percentage, not watts; we still need to confirm it.
- **E2 plan:** Plan: 7 coupons, 2 days and 72 pre-registered samples.
- **Pass rules:** Pre-set pass rules: CV ≤20% and electrical isolation across gaps >3 MΩ.
- **Sealed:** We plan to seal the leads under tape. LIG made in air can absorb water easily[9].

All 35 design-file checks pass. No strip has been cut yet; confirm the LaserCAD import at the machine before fabrication.

Electronics Schematics & BOM

SCHEMATIC CHECK PASSED · NOT YET WIRED



Our full KiCad 10 schematic (reader_v0_1.kicad_sch), ERC 0 violations

This figure shows zone C's measurement path from our KiCad 10 netlist.

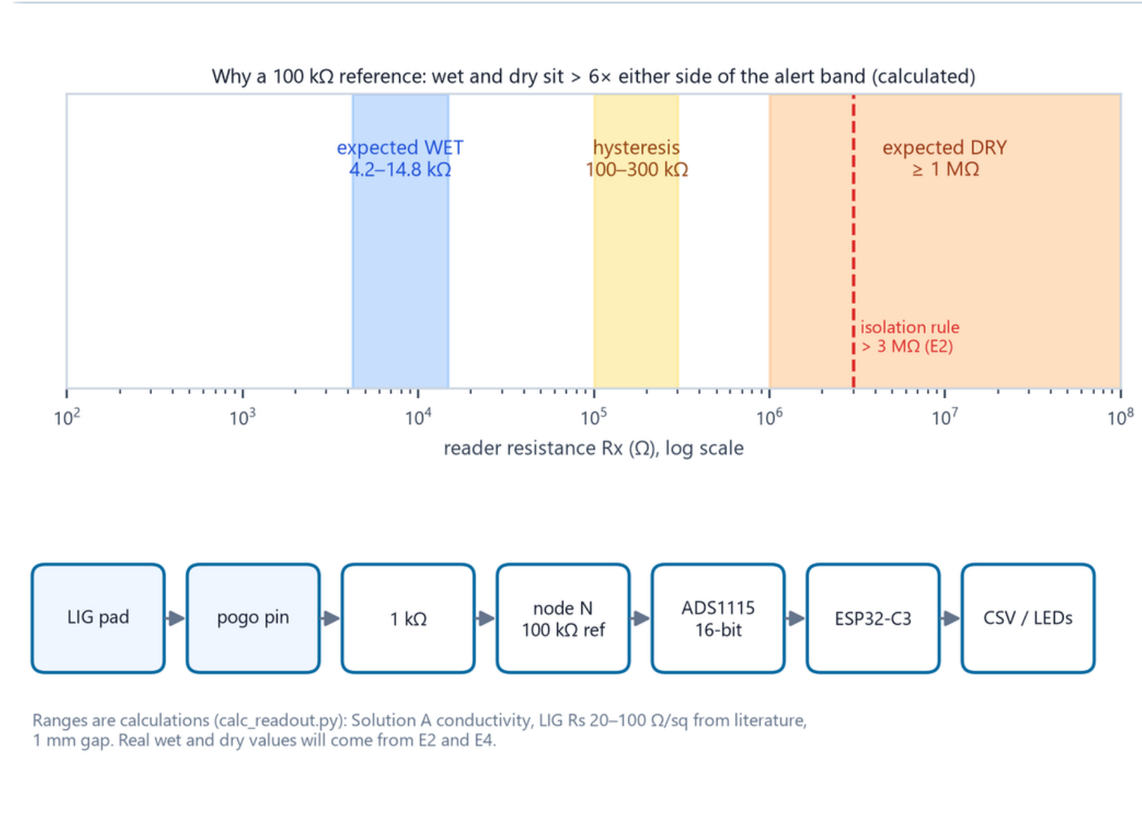
Pin and net map from our KiCad netlist

Net	Pin	Function
DRV_C · DRV_A	GPIO3 · GPIO4	drive zone C · A via 1 kΩ
DRV_LOOP	GPIO5	strip loop (broken strip)
DRV_REF	GPIO10	100 kΩ 0.1 % reference
DRV_CHK	GPIO20	100 kΩ self-check
SDA · SCL	GPIO6 · GPIO7	I²C to ADS1115 (0x48)
N · 3V3	ADS1115 A0 · A1	node · rail (ratiometric)
LED G · Y · R	GPIO0 · 1 · 21	OK · CHANGE · CHECK/FAULT
Strip pads	J4.1 – J4.4	S_C · S_A · COM1 · COM2
Power	USB-C → 3V3	LDO on the SuperMini

The electrical rules check found 0 violations. The netlist matches the firmware and strip pads, with 0 mismatches.

Electronics Schematics & BOM: Bill of Materials

3 ITEMS QUOTED · NOTHING ORDERED



Reusable reader pod: ₹850–1,200 one-time capital estimate. Disposable strip: ₹45–65 each at pilot fabrication scale (estimate).

Bulk prototype purchases and test setup

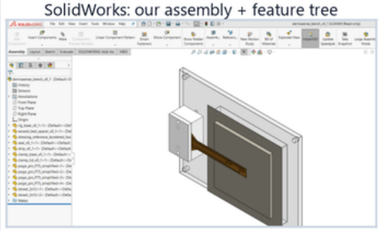
Item	Qty	Source	₹
ESP32-C3 SuperMini (PRO4607)	1	Probots · quoted	242
ADS1115 module (PRO789)	1	Probots · quoted	125
P75-B1 pogo pins (05880)	100	Hubtronics · quoted	354
Resistors, COG cap, LEDs, board	set	local / online	158–421
Clamp + bench rig	set	lab / local	100–1,240
Strip materials	set	lab / online	800–3,500
Test consumables	set	local	480–1,060
Prototype bulk outlay			**≈ 2,259–6,942**

Parts status: Three items quoted on 19 Oct; nothing ordered; other costs are estimates. Total ₹2,259–6,942 covers bulk purchases and bench/test setup, not one strip.

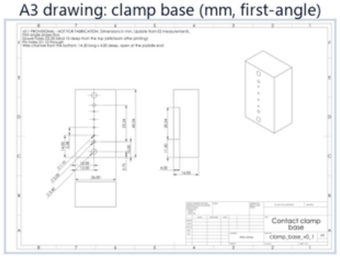
Maker Journey

PROJECT FILES · CHECKS CAN BE RERUN

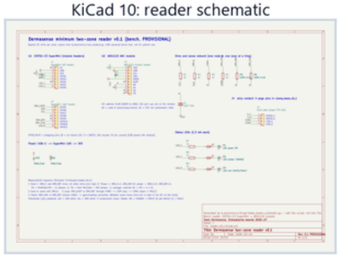
SolidWorks: our assembly + feature tree



A3 drawing: clamp base (mm, first-angle)



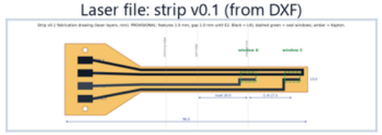
KiCad 10: reader schematic



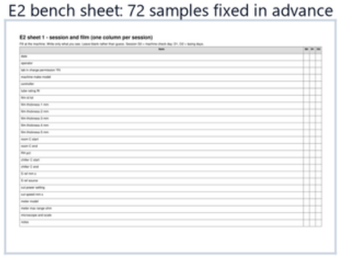
Firmware: state rule + compile-time test

```
// C = pad centre, A = maker's change line
constexpr State classify(Wet c, Wet a,
                         bool fault) {
  return fault ? State::Fault
    : (a == Wet::WET ? (c == Wet::WET
         ? State::Change : State::Check)
       : State::Ok);
}
static_assert(classify(Wet::WET, Wet::WET,
   false) == State::Change,
   "front reached the change line");
```

Laser file: strip v0.1 (from DXF)



E2 bench sheet: 72 samples fixed in advance

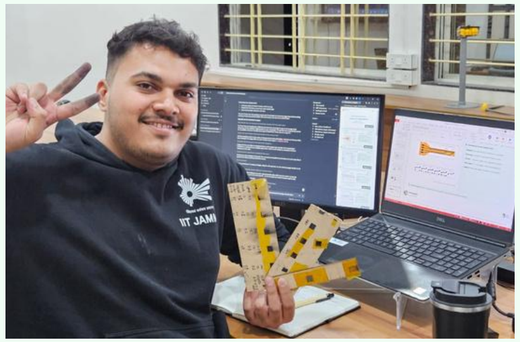


```
Checks re-run on 19 Oct 2026 (results as printed by each tool)

verify_cad.py (SolidWorks)          {'PASS': 99, 'FAIL': 0, 'BLOCKED': 5}
kicad-cli sch erc                   Found 0 violations
verify_laser_files.py               35 PASS, 0 FAIL
arduino-cli compile (esp32c3)       Sketch uses 650349 bytes (49%)
pytest (firmware tools)             16 passed
pytest (E2 analysis, synthetic data) 3 passed
```

Evidence from our project files and checks rerun on 19 October 2026.

Our team and project files



Soham Prabhakar Kadu · role: **[add actual role]**
Bhakti Ajay Kadam · role: **[add actual role]**

LAB PHOTO

the team at work (CAD, wiring or the laser)
replace: right-click → Change Picture

- **Real files:** The repository includes CAD models and drawings, the KiCad schematic, firmware and laser files, with dated work history since 14 October.
- **Re-checkable:** Project scripts can rerun the checks shown here.
- **How we work:** We used Claude to help write some scripts and reviewed their outputs. The physical build and tests are still planned, starting with E2.
- **Photos:** Add authentic team and workspace photos before submission.

Project files: [Add the approved Drive link before submission] · native files and dated work history.

Maker Journey: Engineering Iterations

DECISIONS RECORDED IN OUR PROJECT

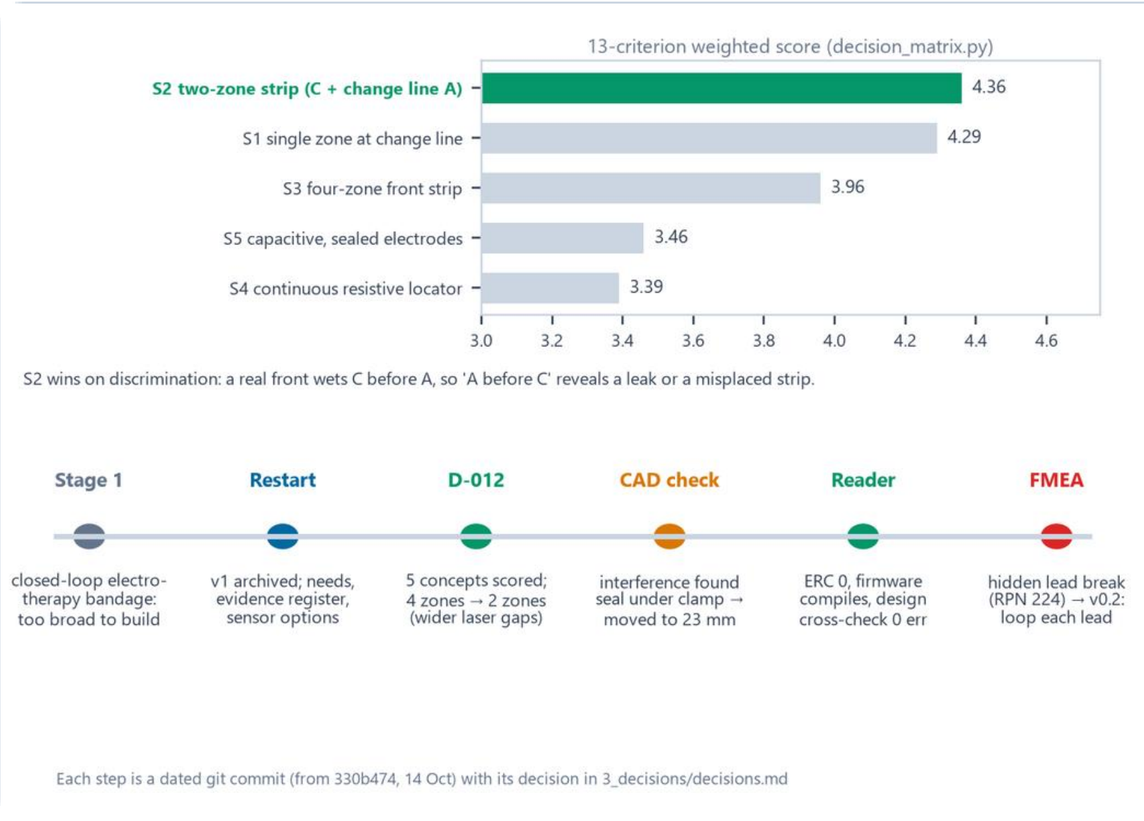


Figure: concept scores and the iteration path (each a dated commit).

How our design changed from Stage 1 to Stage 2

- **Pivot:** We narrowed the idea from a therapy bandage to one question: has the dressing reached its change line?
- **Choice:** The decision matrix slightly favored two zones (4.36 vs 4.29); we kept it to test whether it can flag leakage.
- **Kept:** We kept Bluetooth, but changed it to a local status broadcast. It compiles; hardware testing is still pending.
- **Corrected:** We corrected the Stage 1 description of Kapton: DuPont cautions against calling it medical-grade[11].
- **Redesign:** To address the FMEA risk, version 0.2 loops each sensing lead.

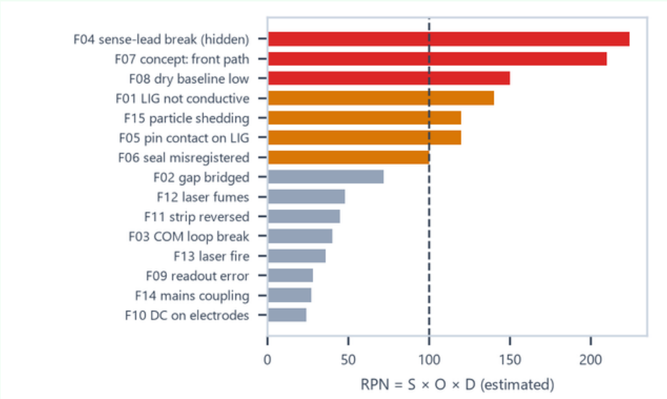
Existing options: diaper strips[4] and visual change indicators[14] already exist. We are testing a two-zone strip that reads the change line electrically and checks which zone wets first.

Safety & Hazard Mitigation

RISKS ESTIMATED · TESTS PLANNED

We ranked 15 possible failure modes before testing. These are the five highest estimated risks.

Failure mode	Effect	RPN	Fix / test
F04 cracked sense lead	A broken sensing lead could make the reader report “dry” continuously.	224	v0.2 loops each lead · T-MECH-01
F07 front path differs	The change alert could be late or absent.	210	E4 on real foam dressings
F08 humid dry baseline	Humidity could trigger a false wet reading.	150	measure baselines (E4)
F01 weak carbon layer	A weak carbon layer could make both zones read dry.	140	E2 + per-strip QC record
F05 pin contact on carbon	Poor pin contact could make readings drift high.	120	rounded pins · T-GEOM-01



Fail-safes

- **Bench only:** Use the USB reader on the bench only; do not put it on a person.
- **No net DC:** The drive is bipolar and idle pins are off. Planned bench test T8 checks for less than 5 mV residual.
- **Self-check:** Run a self-check each scan; show FAULT if it fails.
- **No wound contact:** The strip is designed to sit on the pad’s outer face, with sealed leads.
- **Laser work:** Complete and sign the laser-safety checklist before each session.

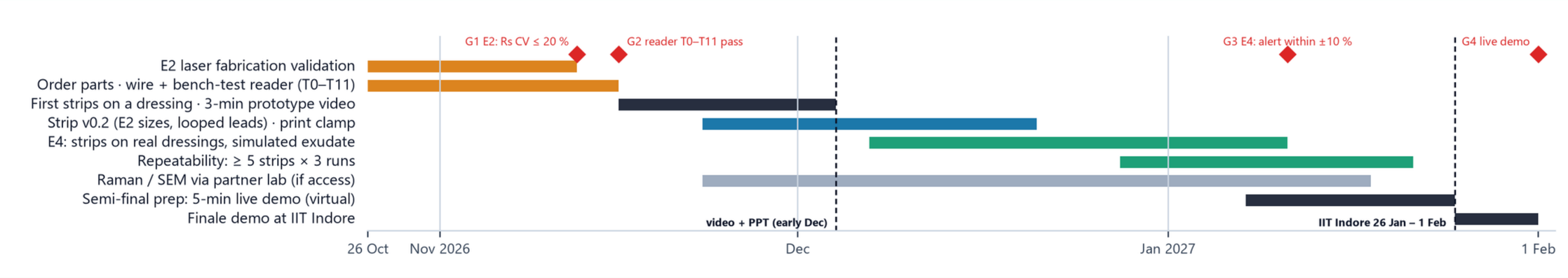
Standards (considered, not claimed)

- **ISO 14971** risk management
- **IEC 60601-1** future on-body reader
- **ISO 10993-1** before any skin contact[11]
- **EN 13726** simulated exudate for tests[10]
- **IEC 60825-1** laser work during fabrication

Roadmap

PLANNED WORK · GATED MILESTONES

Four-month plan aligned with the programme dates. Each step depends on passing the previous gate.



Build & Bench (Months 1 - 2)

- **Nov:** Run E2 laser tests (G1), then build and bench-test the reader (G2).
- **Early Dec:** If bench gates pass, prepare the first dressing demo and 3-minute prototype video.

Validate & Demo (Months 3 - 4)

- **Dec:** Build strip version 0.2 and a printed clamp; run E4 wet tests (G3).
- **Jan:** Test repeatability and prepare a live demo for the virtual semi-final.
- **Stage 3:** Demonstrate the working strip and reader on a dressing, show OK → CHANGE, and save the raw data.

Technical Challenges & Risks

ISSUES FOUND DURING OUR CHECKS

Geometry: seal tape under the clamp

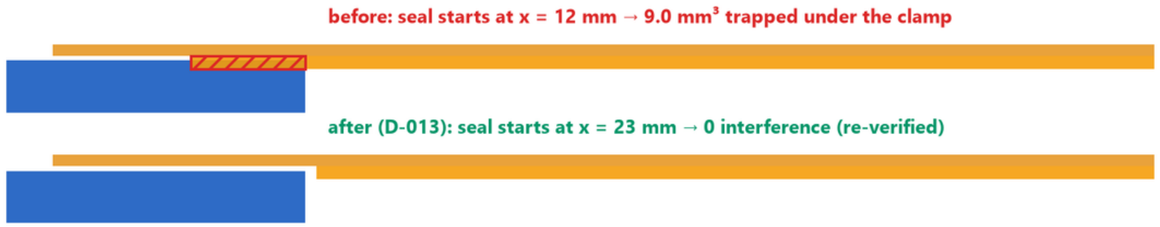
Found: We found the seal tape overlapped the clamp by 9.0 mm³.
Impact: This could lift the strip off the contact pins.
How: We found it with an automated CAD clash check.

Circuit: a broken lead can look dry

Found: A cracked sensing lead could make the reader report “dry”.
Impact: The CHANGE status might never appear, so there would be no alert.
How: The FMEA ranked this risk at 224.

Process: laser setting still unconfirmed

Found: The machine’s “12” setting is likely a PWM percentage. Laser output can be unstable at low power[8].
Impact: Strip dimensions remain provisional until the laser setting is confirmed.
How: We found the uncertainty while checking the LaserCAD configuration.



Side section along the strip (thickness exaggerated): clamp base (blue), strip (amber), seal tape (orange)



We also corrected two drawing issues: STEP files now export in millimetres, and a 0.125 mm dimension is shown correctly.

Potential Solutions & Risk Mitigation

CHECKED · DESIGNED · TO TEST

Checked in CAD or software

Seal: starts at x = 23 mm; 0 clash.
Units: all STEP files in mm.
Checks: 99 checks passed, 0 failed; 5 await E2 measurements.

Designed, not yet tested

v0.2 strip: every sense lead looped.
Contact: silver pads, rounded pins.
Self-test: check resistor → FAULT.

Fallbacks

Wider lines: if E2 says so.
Process: slower passes, log current.
Concept: one-zone strip (score 4.29).

What we still need to test

- **Experiments:** Run E2 laser tests, bench-test the reader, then run E4 wet tests.
- **Material:** Use Raman and SEM before describing the material as graphene.
- **Reliability:** Test bending, contact wear and humidity-related drift.
- **Product:** Explore a battery-powered reader and a secure local phone connection. Neither is built yet; one strip is intended per dressing.
- **Before any study with people: address relevant biocompatibility and electrical-safety requirements and obtain ethics approval.**
- **Not claimed:** We have not established sensor performance, graphene identity or clinical benefit.

Our rule: we will not claim sensor performance, graphene identity or clinical benefit until tests support those claims.

References & Sources

SOURCES AND WHAT WE READ

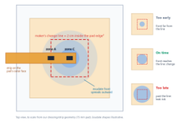
The superscript numbers point to sources below. We note whether we read the full paper, an abstract, a summary or a product page; a citation alone does not mean a claim is independently verified.

1. Milne S.D. et al., A wearable wound moisture sensor as an indicator for wound dressing change, Int. Wound J. 13(6):1309–1314 (2016). doi:10.1111/iwj.12521 · full text read
2. Dressing instructions for use: some foam dressings say 'change when exudate approaches ~2 cm from the pad edge'; Mölnlycke Mepilex Border Flex brochure uses strike-through reaching 3 edges · manufacturer documents, read as excerpts
3. McColl D., Cartlidge B., Connolly P., Real-time monitoring of moisture levels in wound dressings in vitro, Int. J. Surg. 5(5):316–322 (2007). doi:10.1016/j.ijsu.2007.02.008 · abstract
4. US 5,469,145 and US 5,838,240 (wet diaper detector: detachable strip + clip-on detector); US 9,291,589; US 10,928,344 · patent summaries, claims not read
5. Jiang Y. et al., Wireless, closed-loop, smart bandage with integrated sensors and stimulators, Nat. Biotechnol. (2022). doi:10.1038/s41587-022-01528-3 · listing + coverage
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7. Lin J. et al., Laser-induced porous graphene films from commercial polymers, Nat. Commun. 5:5714 (2014). doi:10.1038/ncomms6714 · full text read
8. CO_2 laser tube current vs PWM setting, engineering measurements, softsolder.com (2022) · not peer-reviewed
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10. EN 13726-1:2002 / EN 13726:2023, Test methods for wound dressings (absorbency; 'Solution A') · catalogue entry; online link not reachable from our network
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17. Sharma A. et al., Epidemiological profile of chronic wounds in an Indian population, Cureus 16(9):e68684 (2024). doi:10.7759/cureus.68684 · full text
18. Anjana R.M. et al., ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17), Lancet Diabetes Endocrinol. 11(7):474–489 (2023) · abstract and summaries
19. Armstrong D.G., Boulton A.J.M., Bus S.A., Diabetic foot ulcers and their recurrence, N. Engl. J. Med. 376:2367–2375 (2017). doi:10.1056/NEJMra1615439 · summaries; range is an estimate and contested
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Where the figures came from

FIGURES MADE FROM OUR PROJECT FILES

The illustrations come from our CAD, schematic exports and plots made from project files. We did not use stock images or AI image generators. Team photos still need to be added by the team.



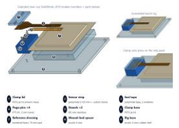
Problem illustration, drawn to scale
Source: make_figures.py · strip_v0_1_geometry.json
1_presentation/figures/fig_problem.png



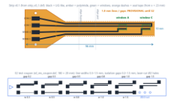
How it senses (concept)
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1_presentation/figures/fig_howitworks.png



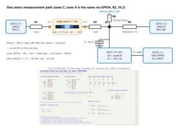
Firmware scan and state logic
Source: make_figures.py · reader_v0_1 source
1_presentation/figures/fig_dataflow.png



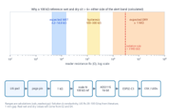
Annotated SolidWorks views
Source: render_cad.py + make_figures.py · 5_cad/solidworks
1_presentation/figures/fig_cad.png



Strip and E2 coupon layers
Source: make_figures.py · our DXF files
1_presentation/figures/fig_strip.png



Measurement path + KiCad sheet
Source: make_figures.py · reader_v0_1 netlist + KiCad PDF
1_presentation/figures/fig_schematic.png



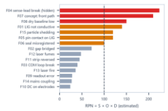
Measurement window
Source: make_figures.py · calc_readout.py
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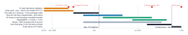
Evidence gallery + tool output
Source: our files, screen capture, 19 Oct tool runs
1_presentation/figures/fig_maker.png



Decision matrix + iterations
Source: make_figures.py · decision_matrix.md
1_presentation/figures/fig_decision.png



FMEA risk ranking
Source: make_figures.py · fmea_v0_1.csv
1_presentation/figures/fig_fmea.png



4-month roadmap
Source: make_figures.py · E2/bench/E4 plans
1_presentation/figures/fig_roadmap.png



Seal clash + lead-break diagrams
Source: make_figures.py · geometry + FMEA
1_presentation/figures/fig_challenges.png